

ADVANCED PROGRAMMING PRACTICE (21CSC203P)

Unit – 1 : Session – 11 : SLO – 1 | High-level to Machine Code

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Activity (Solutions)

- 1) Compile + disassemble: `javac Hello.java` produces `Hello.class`; `javap -c Hello` shows the bytecode, e.g.,
main: 0: `getstatic #2 (System.out)` 3: `ldc #3 ("Hello")` 5: `invokevirtual #4 (println)` 8: `return` — mnemonics the JVM executes.
- 2) JVM bytecode → machine code: the interpreter executes bytecode instruction by instruction; the JIT (Just-In-Time) compiler detects hot spots (frequently run code) and compiles them to native machine code, which the CPU then executes directly.
- 3) Class loaders: Bootstrap loader loads core JDK classes (`java.lang.*`) from the runtime; Extension/Platform loader loads extension libraries; Application/System loader loads classes from the application classpath. They work in a delegation hierarchy — each loader first asks its parent, ensuring core classes cannot be overridden and classes load once.